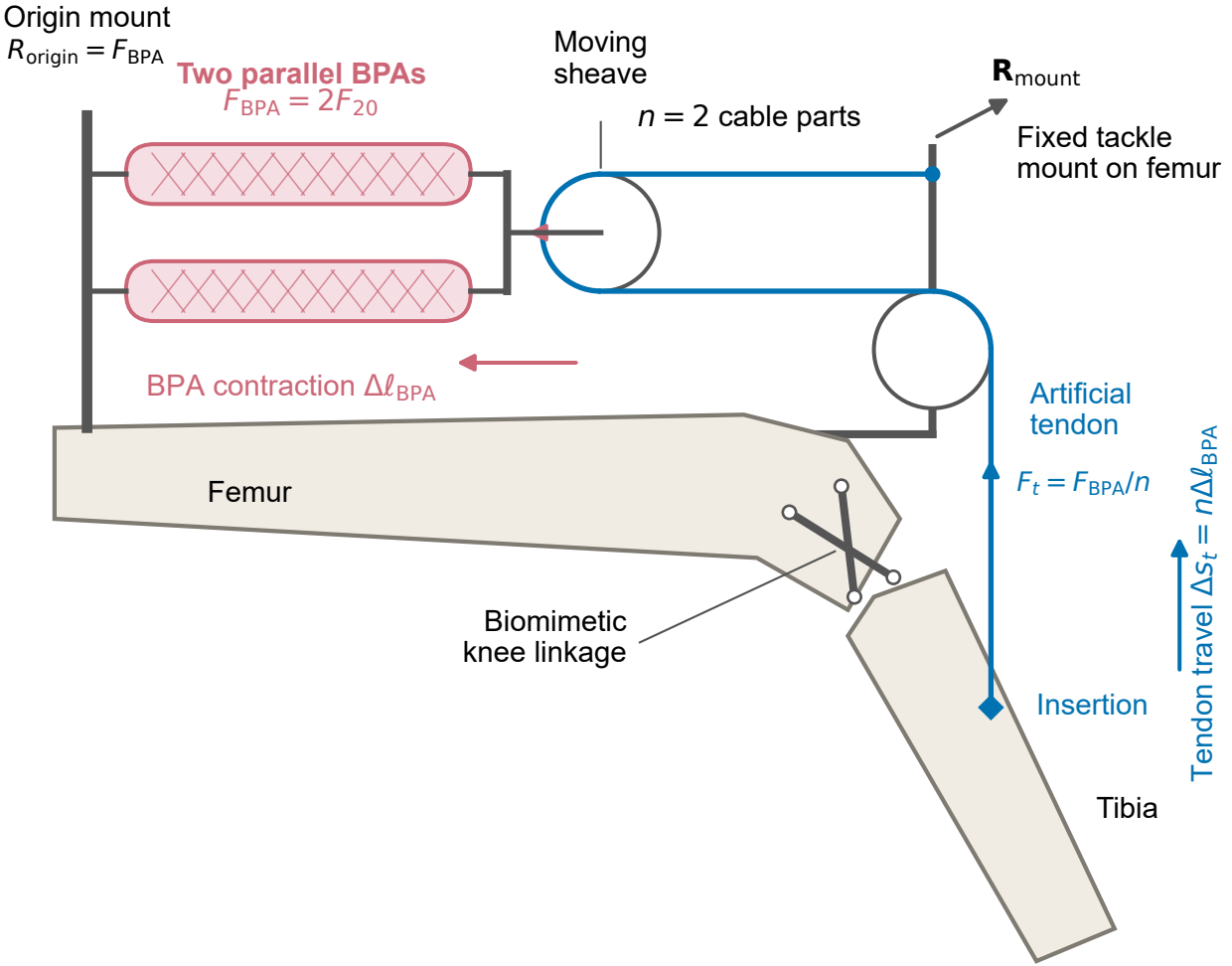
Origin mount
$R_{origin} = F_{BPA}$
Two parallel BPAs
$F_{BPA} = 2F_{20}$
Moving sheave
$n = 2$ cable parts
$\mathbf{R}_{mount}$
Fixed tackle mount on femur
BPA contraction $\Delta\ell_{BPA}$
Artificial tendon
$F_t = F_{BPA}/n$
Femur
Biomimetic knee linkage
Insertion
Tibia
Tendon travel $\Delta s_t = n\Delta\ell_{BPA}$